Green Customer Insights Platform

1 Challenge

Build an AI platform that analyses customer transactions, financial products and behavioural patterns to:

- identify opportunities for green financial products;
- estimate the potential sustainability impact;
- generate personalised sustainability insights for customers;
- give actionable insights to business and relationship teams;
- encourage paperless and digital journeys;
- explain why each opportunity was identified.

Core rule. ML models make predictions. A rules-based carbon engine computes impact using documented formulas, emissions factors and assumptions. The LLM only explains structured outputs; it does not make predictions, calculate impact, or decide financial eligibility. Financial eligibility is determined by product-specific rules and the institution's authorised processes.

2 Green Opportunities

Opportunities 1 to 4 are required. Opportunities 5 and 6 are optional.

#	Opportunity	Example customer signal	Product
1	EV Finance	High recurring fuel spend, existing vehicle loan	GP01 EV Finance
2	Green Home	Mortgage or home ownership, high utility spend	GP02 Green Home Loan
3	Solar Finance	Home ownership, high electricity spend	GP03 Solar Finance
4	Paperless Banking	Paper statements, high digital activity	GP04 Paperless Banking
5	Green Investments	Existing investments, sustainability preference	Optional
6	Green Loans	Existing borrowing, eligible use case	Optional

The customer signals are illustrative and may overlap. The green-products catalogue should define the intended use cases for Green Home and Solar Finance so recommendations are mapped consistently. A signal or propensity does not establish financial eligibility.

3 Key Terms

Term	Definition
Cutoff date	The last date covered by the input data.
Take-up	The customer adopts the product within a fixed outcome window after the cutoff date; teams must state the window used.
Propensity	Estimated probability of take-up, in [0, 1].
Confidence	How reliable the system considers a propensity estimate, in [0, 1]. The method is the team's choice and must be documented. It is distinct from propensity and does not indicate financial eligibility.
Reason code	A fixed identifier for one factor behind a recommendation, e.g. `HIGH_FUEL_SPEND`.
Carbon engine	A deterministic, rules-based component that calculates estimated emissions or emissions avoided from activity data, documented formulas and baselines, and emissions factors. It does not make behavioural predictions or determine eligibility.
Financial eligibility	Whether a customer qualifies under the institution's authorised product criteria and processes; propensity is not eligibility.
Estimate	A modelled potential value, not a measured result. All impact figures are estimates.

4 Dataset

Teams are expected to prepare the data needed for their solution. They may generate synthetic data, use suitable publicly available datasets that may be lawfully used, or combine both. Real personal or confidential banking data must not be used. The tables below suggest entities and fields; teams may adapt them, provided the data supports all required opportunities and functional requirements. Teams must document their data sources, schema, assumptions and limitations, along with the generation process and label-generation rules for synthetic data. Where customer-level data is used, teams should use customer-level train, validation and test splits and avoid data leakage between splits.

Table	Main fields
Customers	customer_id, age, income_band, postcode, home_owner, digital_usage, sustainability_preference
Transactions	transaction_id, customer_id, date, merchant_description (raw text), amount, debit_credit
Products	customer_id, product, start_date, balance, status
Vehicles	customer_id, vehicle_type, fuel_type, vehicle_age, annual_mileage
Energy usage	customer_id, month, electricity_kwh, gas_kwh
Digital usage	customer_id, month, digital_statements, paper_statements, digital_transactions, branch_visits
Green products	product_id, product, category, eligibility, sustainability_metric
Emissions factors	activity, unit, kg_co2e_per_unit, source
Labels	transaction category; take-up (0/1) per customer and product

Teams must provide or construct the transaction-category and customer-product take-up labels needed to train and evaluate their models, and document how labels were obtained or generated. Synthetic labels must reflect plausible relationships with input features rather than being assigned arbitrarily.

Teams must use documented emissions factors for all carbon estimates, including units and sources. Where factors are not available in their data, teams must source appropriate factors and include them with the submission.

Suggested public datasets (optional). Teams may find the following datasets useful:

- **UCI Online Retail II:** Dataset link. Contains customer identifiers, transaction dates, product descriptions, quantities and prices. It can be used for transaction analysis and customer-level feature engineering.
- **London Smart Meter Energy Consumption:** Dataset link. Contains household electricity consumption readings and can be used to develop energy-use profiles for Green Home and Solar Finance opportunities.

5 Functional Requirements

FR1 Customer 360. Combine all tables into one view per customer.

FR2 Transaction categorisation. Assign each transaction exactly one sustainability category (Fuel, Electricity, Gas, Public Transport, Home Improvement, Automotive, Utilities, Other). Categories must be mutually exclusive; teams must document boundary decisions (e.g. Fuel vs Automotive, or Electricity/Gas vs Utilities) and apply them consistently. Rules, ML or embeddings may be used.

FR3 Opportunity detection. For each customer and required product, output a propensity, confidence and reason codes.

FR4 Sustainability impact. Estimate potential impact, for example current transport emissions against emissions after moving to an EV. Document assumptions, baselines and time horizons. Use an appropriate documented methodology (e.g. relevant GHG Protocol guidance for project or avoided-emissions estimates, or PCAF for financed-emissions accounting) rather than inventing a method.

FR5 Customer insight. Generate a customer-friendly explanation from the structured outputs only.

FR6 Business insights. Show high-potential segments, propensity scores, regional and trend insights, conversion opportunities and estimated impact.

FR7 Explainability. Every recommendation states why it was made: evidence, confidence, reason codes and data timestamp. Products the customer's data rules out (e.g. EV finance for an EV owner where no replacement use case is present) must not be recommended.

Example output:

```
customer_id:   C10001
opportunity:   EV_FINANCE
propensity:    0.84
confidence:    0.87
reason_codes:  HIGH_FUEL_SPEND, HIGH_ANNUAL_MILEAGE, EXISTING_VEHICLE
```

6 Architecture

Pipeline. Data → Feature store → Transaction classification, propensity models, deterministic carbon engine → Recommendation engine → LLM insight layer → Dashboard and API.

Example of the role split. The ML model determines "82% propensity for EV finance". The carbon engine determines "estimated annual reduction of X kg CO_2e". The LLM turns these into a plain-language explanation.

Fine-tuning is optional.

Permitted stack. Python, Pandas, NumPy, SQL, PostgreSQL, SQLite, PySpark; scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow; OpenAI, Azure OpenAI, Gemini, Anthropic, Hugging Face, Llama and Mistral models; FAISS, Chroma, Qdrant, Pinecone; FastAPI, Flask; Streamlit (preferred), React, Next.js; Plotly, Power BI, Tableau.

7 User Experience

Customer view. Green insights, relevance, evidence, potential annual impact (labelled as an estimate, with the method) and a clear action, such as *Explore EV options* or *Go paperless*.

Business dashboard. Customers analysed, opportunities by product, potential impact (CO_2e, energy, paper, digital adoption) and top segments, with drill-down: Customer $\rightarrow$ Opportunity $\rightarrow$ Evidence $\rightarrow$ Impact $\rightarrow$ Action. Report emissions, energy, paper and digital-adoption metrics separately; digital adoption is not itself a carbon reduction.

8 Optional Extension: Customer Memory

Teams may give each customer a persistent memory, such as past insights and the customer's responses, and use it to adapt later recommendations. The design is open. Propensity must come from the ML models, eligibility must follow the institution's authorised product rules and processes, and impact must come from the carbon engine. Customer memory and the LLM must not override these outputs. Teams that attempt this should show how recommendations change as memory builds up.

9 Evaluation

Teams must evaluate their solutions using documented train, validation and test splits where applicable. They must report their evaluation methodology, relevant metrics, baseline comparisons, and limitations. The quality of the data, label-generation process, validation methodology and reproducibility will be considered alongside reported results.

Since teams may use different datasets, numerical metrics from different submissions are not directly comparable and will not be used as a standalone ranking criterion.

Area	Evaluation criteria
Data quality	Plausibility of the data, documented sources and assumptions, label-generation methodology, and prevention of data leakage.
ML and recommendations	Appropriateness of the approach, validation methodology, comparison with a simple baseline, and quality of the resulting recommendations.
Transaction categorisation	Consistency and correctness of categorisation, supported by appropriate evaluation metrics and examples.
Sustainability impact	Correctness and reproducibility of calculations, use of documented emissions factors, and transparency of assumptions.
LLM explanations	Groundedness, factual accuracy, completeness, correct product mapping, and appropriate handling of missing or conflicting information.
System performance	Responsiveness, reliability, reproducibility and reasonable computational and LLM costs.

Scenario-based evaluation. During evaluation, organisers may test submissions using scenarios involving missing data, conflicting signals, products that should not be recommended, and insufficient information to estimate sustainability impact.

Baseline and reproducibility. Teams must compare their approach with a simple baseline on their own validation data and explain the results. Organisers may rerun submitted code and inspect the data, calculations and outputs. Reported results that cannot be reproduced may affect the evaluation.

10 Submission

A. Working prototype (web app or dashboard)

B. Source code, with run instructions
C. Trained models and inference code
D. Preprocessing pipeline
E. LLM implementation: prompts, model or API details, evaluation approach
F. Architecture diagram
G. Evaluation report: the metrics in Section 9, on the validation set
H. Responsible AI note: privacy, bias, explainability, hallucination, human oversight, data limitations
I. Test set predictions

11 Scoring

Criterion	Weight
Business relevance	15%
ML and recommendation quality (data quality, validation methodology and held-out metrics)	20%
Sustainability insight quality	15%
GenAI and LLM implementation (evaluation set)	15%
Technical architecture (including data engineering and system performance)	15%
Explainability and Responsible AI	10%
UX and dashboard	5%
Innovation (including the optional extension)	5%
Total	100%